

Reasoning Behavior Analysis Report

EXPERIMENTAL SETUP

=====

Data Folder: data/MATH500/deepseek/p2
Total Items: 500
Total Strategies: 9

OVERALL ACCURACY

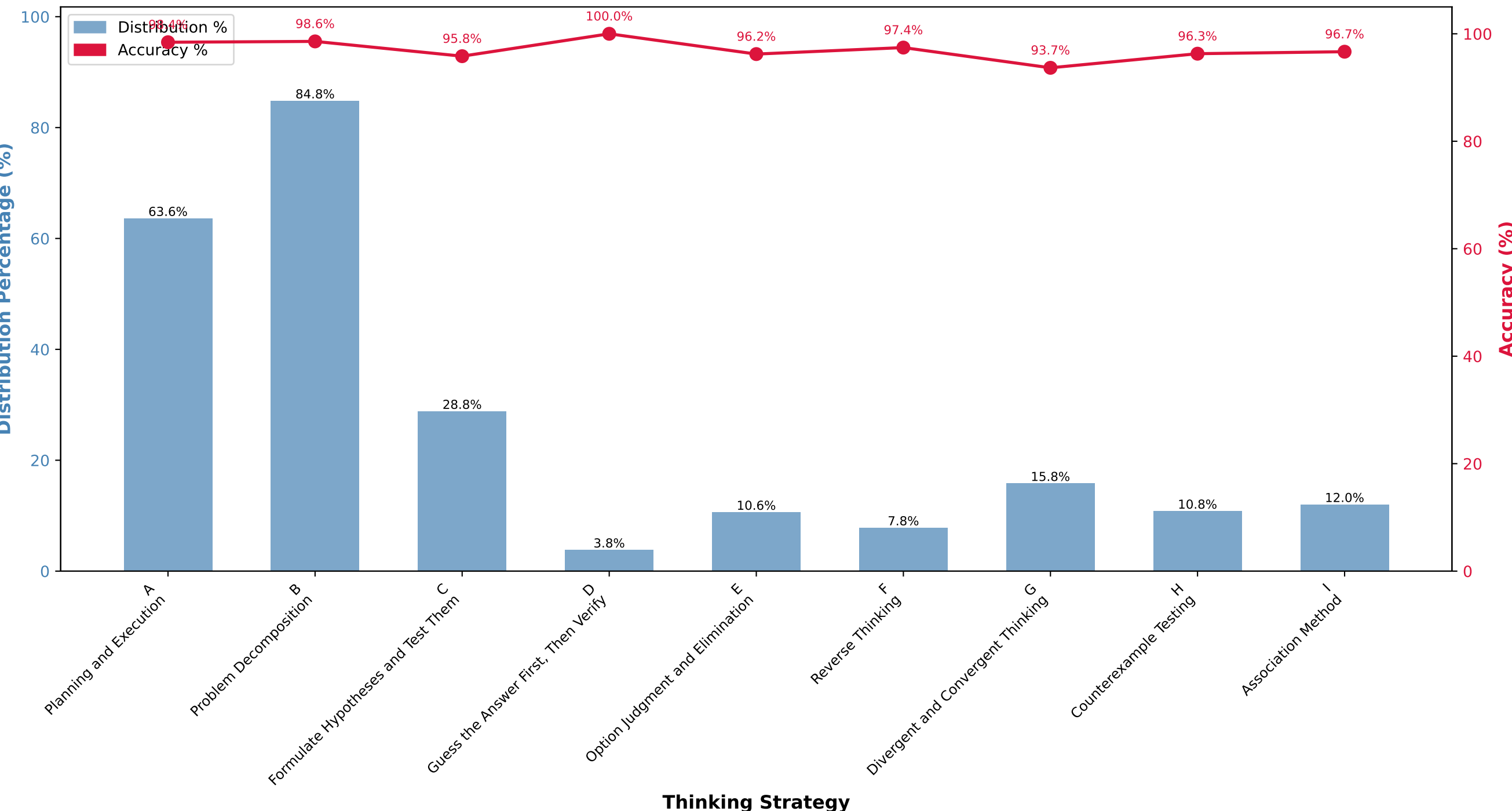
=====

Total Problems: 500
Correct: 494
Incorrect: 6
Overall Accuracy: 98.80%

THINKING STRATEGY DISTRIBUTION

- =====
- A: Planning and Execution
Distribution: 63.6%
Accuracy: 98.43% (313/318)
 - B: Problem Decomposition
Distribution: 84.8%
Accuracy: 98.58% (418/424)
 - C: Formulate Hypotheses and Test Them
Distribution: 28.8%
Accuracy: 95.83% (138/144)
 - D: Guess the Answer First, Then Verify
Distribution: 3.8%
Accuracy: 100.00% (19/19)
 - E: Option Judgment and Elimination
Distribution: 10.6%
Accuracy: 96.23% (51/53)
 - F: Reverse Thinking
Distribution: 7.8%
Accuracy: 97.44% (38/39)
 - G: Divergent and Convergent Thinking
Distribution: 15.8%
Accuracy: 93.67% (74/79)
 - H: Counterexample Testing
Distribution: 10.8%
Accuracy: 96.30% (52/54)
 - I: Association Method
Distribution: 12.0%
Accuracy: 96.67% (58/60)

Thinking Strategy Distribution and Accuracy



Example for Strategy A: Planning and Execution

PROBLEM

Let $f(x) = (\arccos x)^2 + (\arcsin x)^2$. Find the range of $f(x)$. All functions are in radians.

GROUND TRUTH ANSWER

$\left[\frac{\pi^2}{8}, \frac{5\pi^2}{4} \right]$

MODEL RESPONSE

The function $f(x) = (\arccos x)^2 + (\arcsin x)^2$ is defined for $x \in [-1, 1]$.

Using the identity $\arcsin x + \arccos x = \frac{\pi}{2}$, let $u = \arcsin x$, so $\arccos x = \frac{\pi}{2} - u$. Then $f(u) = \left(\frac{\pi}{2} - u \right)^2 + u^2 = 2u^2 - \pi u + \frac{\pi^2}{4}$, where $u \in \left[-\frac{\pi}{2}, \frac{\pi}{2} \right]$.

This is a quadratic function in u with a minimum at $u = \frac{\pi}{4}$ (since the vertex is at $u = \frac{\pi}{4}$, which lies within the interval). The values at the critical points are:
- At $u = \frac{\pi}{4}$, $f\left(\frac{\pi}{4}\right) = \frac{\pi^2}{8}$.
- At $u = -\frac{\pi}{2}$, $f\left(-\frac{\pi}{2}\right) = \frac{5\pi^2}{4}$.

CHAIN OF THOUGHT (First 1500 chars)

First, the function is $f(x) = (\arccos x)^2 + (\arcsin x)^2$. I need to find its range, meaning all possible values that $f(x)$ can take for x in its domain.

I recall that $\arcsin$ and $\arccos$ have specific domains and ranges. $\arcsin x$ is defined for x in $[-1, 1]$ and its range is $[-\pi/2, \pi/2]$. Similarly, $\arccos x$ is defined for x in $[-1, 1]$ and its range is $[0, \pi]$.

So, the domain of $f(x)$ is $x \in [-1, 1]$.

Now, I need to find the sum of squares of these two functions.

I remember that $\arcsin x$ and $\arccos x$ are related. Specifically, for all x in $[-1, 1]$, we have:

$$\arcsin x + \arccos x = \frac{\pi}{2}$$

Is that true? Let me verify. Yes, because $\sin(\arcsin x) = x$ and $\cos(\arccos x) = x$, and since $\sin(\theta) = \cos(\pi/2 - \theta)$, it follows that $\arcsin x = \pi/2 - \arccos x$, so indeed:

$$\arcsin x + \arccos x = \frac{\pi}{2}$$

That's a key identity.

So, let me set $u = \arcsin x$. Then $\arccos x = \pi/2 - u$, since $\arcsin x + \arccos x = \pi/2$.

$$\text{Then } f(x) = (\arccos x)^2 + (\arcsin x)^2 = (\pi/2 - u)^2 + u^2$$

Where $u = \arcsin x$, and since $x \in [-1, 1]$, $u \in [-\pi/2, \pi/2]$.

$$\text{So, } f(u) = (\pi/2 - u)^2 + u^2$$

Now, I can simplify this expression.

$$\text{Expand it: } f(u) = (\pi/2)^2 - 2(\pi/2)u + u^2 + u^2 = (\pi^2/4) - \pi u + 2u^2$$

$$\text{So, } f(u) = 2u^2 - \pi u + \pi^2/4$$

This is a quadratic function in u . Since the coefficient of u^2 is positive ($2 > 0$), it opens upwards, so it has a minimum at the vertex.

The vertex of a quadratic $au^2 + bu + c$ is at $u = -b/(2a)$

Here, $a = 2$, $b = -\pi$, so $u_{\text{vertex}} = -(-\pi)/(2*2) = \pi/4$.

Example for Strategy B: Problem Decomposition

PROBLEM

Find the greatest common divisor of \$3339\$, \$2961\$, and \$1491\$.

GROUND TRUTH ANSWER

21

MODEL RESPONSE

The greatest common divisor (GCD) of 3339, 2961, and 1491 is 21. This is determined using the Euclidean algorithm:

- GCD(3339, 2961) = 63
 - 3339 ÷ 2961 = 1 with remainder 378
 - GCD(2961, 378) = 315
 - 2961 ÷ 378 = 7 with remainder 315
 - GCD(378, 315) = 63
 - 378 ÷ 315 = 1 with remainder 63
 - GCD(315, 63) = 63
 - 315 ÷ 63 = 5 with remainder 0
- GCD(63, 1491) = 21
 - 1491 ÷ 63 = 23 with remainder 42
 - GCD(63, 42) = 21
 - 63 ÷ 42 = 1 with remainder 21
 - GCD(42, 21) = 21
 - 42 ÷ 21 = 2 with remainder 0

Thus, GCD(3339, 2961, 1491) = 21. Verification shows that 21 divides all three numbers evenly:

- 3339 ÷ 21 = 159
- 2961 ÷ 21 = 141
- 1491 ÷ 21 = 71

CHAIN OF THOUGHT (First 1500 chars)

First, I need to find the greatest common divisor (GCD) of three numbers: 3339, 2961, and 1491. The GCD is the largest number that divides all three without leaving a remainder.

I should use the Euclidean algorithm, which is efficient for finding GCDs. Since there are three numbers, I can find the GCD of two first and then find the GCD of that result with the third number.

So, let me start with two numbers, say 3339 and 2961.

I need to compute GCD(3339, 2961).

Using the Euclidean algorithm:

Divide the larger number by the smaller number and find the remainder.

3339 ÷ 2961 = 1 with a remainder? Let me calculate.

$$2961 \times 1 = 2961$$

$$3339 - 2961 = 378$$

So, remainder is 378.

Now, GCD(2961, 378) because $\text{GCD}(a, b) = \text{GCD}(b, a \bmod b)$

So, now find GCD(2961, 378)

Divide 2961 by 378.

$$378 \times 7 = 2646$$

$$2961 - 2646 = 315$$

So, remainder is 315.

Now, GCD(378, 315)

Divide 378 by 315.

$$315 \times 1 = 315$$

$$378 - 315 = 63$$

So, remainder is 63.

Now, GCD(315, 63)

Divide 315 by 63.

$$63 \times 5 = 315$$

Exactly, so remainder is 0.

Therefore, $\text{GCD}(315, 63) = 63$

So, $\text{GCD}(3339, 2961) = 63$

Now, I need to find GCD of this result with the third number, 1491.

So, find GCD(63, 1491)

Now, since 63 is smaller, I divide 1491 by 63.

Let me calculate $1491 \div 63$.

First, $63 \times 23 = ?$ $60 \times 23 = 1380$, $3 \times 23 = 69$, so $1380 + 69 = 1449$

$$1491 - 1449 = 42$$

So, remainder is 42.

Now, GCD(63, 42)

Divide 63 by 42.

$$42 \times 1 = 42$$

$$63 - 42 = 21$$

Remainder is 21.

Now, GCD(42, 21)

Divide 42 by 21.

$$21 \times 2 = 42$$

Exactly, remainder 0.

...

Example for Strategy C: Formulate Hypotheses and Test Them

PROBLEM

A 90° rotation around the origin in the counter-clockwise direction is applied to $7 + 2i$. What is the resulting complex number?

GROUND TRUTH ANSWER

$-2 + 7i$

MODEL RESPONSE

A 90° counter-clockwise rotation around the origin in the complex plane corresponds to multiplication by the imaginary unit i .

Given the complex number $7 + 2i$, the rotation is computed as follows:

```
\\[
i \\times (7 + 2i) = 7i + 2i^2 = 7i + 2(-1) = 7i - 2 = -2 + 7i
\\]
```

Thus, the resulting complex number after the rotation is $-2 + 7i$.

CHAIN OF THOUGHT (First 1500 chars)

First, the user is asking about a 90° rotation around the origin applied to the complex number $7 + 2i$. I need to find the resulting complex number after this rotation.

I recall that in the complex plane, a rotation by 90° counter-clockwise corresponds to multiplication by the imaginary unit i . That's because multiplying by i rotates a complex number by 90° in the counter-clockwise direction.

So, if I have a complex number $z = 7 + 2i$, then after a 90° counter-clockwise rotation, the new number should be $i \times z$.

Let me compute that: $i \times (7 + 2i) = i \times 7 + i \times 2i = 7i + 2i^2$.

I know that $i^2 = -1$, so $2i^2 = 2 \times (-1) = -2$.

Therefore, $7i + (-2) = -2 + 7i$.

So, the resulting complex number should be $-2 + 7i$.

I should double-check. The original point is at $(7, 2)$ in the complex plane, where the x-axis is real and y-axis is imaginary. A 90° counter-clockwise rotation would map (x, y) to $(-y, x)$. That's a standard rotation.

So, for $(7, 2)$, after rotation, it should be $(-2, 7)$, which corresponds to the complex number $-2 + 7i$. That matches what I got.

Therefore, the answer is $-2 + 7i$.

The user said "a 90° rotation around the origin in the counter-clockwise direction", which is exactly what I did.

I think that's correct.

Example for Strategy D: Guess the Answer First, Then Verify

PROBLEM

Solve for x : $3^{2x} + 19 = 10^x$.

GROUND TRUTH ANSWER

2

MODEL RESPONSE

The equation to solve is $3^{2x} + 19 = 10^x$.

Notice that $3^{2x} = (3^2)^x = 9^x$, so the equation simplifies to:

$$9^x + 19 = 10^x$$

Rearranging terms:

$$10^x - 9^x = 19$$

By inspection, try $x = 2$:

$$10^2 - 9^2 = 100 - 81 = 19$$

Thus, $x = 2$ is a solution.

Consider the function $f(x) = 10^x - 9^x$. For $x < 0$, $10^x < 9^x$ because $10 > 9$, so $f(x) < 0$. For $x = 0$, $f(0) = 1 - 1 = 0$. For $x > 0$, $f(x)$ is strictly increasing because the derivative $f'(x) = 10^x \ln 10 - 9^x \ln 9 > 0$ for $x > 0$ (since $10^x > 9^x$ and $\ln 10 > \ln 9$).

Therefore, $f(x) = 19$ only at $x = 2$, and there are no other solutions.

$$\boxed{2}$$

CHAIN OF THOUGHT (First 1500 chars)

First, the equation is $3^{2x} + 19 = 10^x$. I need to solve for x . This looks tricky because it involves exponential terms with different bases. I should try to simplify it or see if there's a way to rewrite it.

Notice that $3^{2x} = (3^2)^x = 9^x$. So the equation becomes $9^x + 19 = 10^x$.

Now, I have $10^x - 9^x = 19$. That might be a bit easier to work with.

I recall that for exponential equations, sometimes we can take logarithms, but here the bases are different, so that might not directly help. Alternatively, I might consider if there's an obvious solution by inspection.

Let me try plugging in some values for x .

If $x = 0$, $10^0 - 9^0 = 1 - 1 = 0$, which is not 19.

If $x = 1$, $10^1 - 9^1 = 10 - 9 = 1$, not 19.

If $x = 2$, $10^2 - 9^2 = 100 - 81 = 19$. Oh, that works! So $x = 2$ is a solution.

But is it the only solution? I should check if there are other solutions.

Let me think about the function $f(x) = 10^x - 9^x$. I need to see where this equals 19.

I know that for $x < 0$, both 10^x and 9^x are positive, but $10^x < 9^x$ because $10 > 9$, so for negative x , $10^x - 9^x < 0$, so it can't be 19 since 19 is positive.

For $x = 0$, it's 0, and as x increases, 10^x grows faster than 9^x , so $f(x) = 10^x - 9^x$ increases. Let me check the derivative to confirm.

Set $f(x) = 10^x - 9^x$. Then $f'(x) = 10^x \ln 10 - 9^x \ln 9$.

Since $10^x > 9^x$ for $x > 0$...

Example for Strategy E: Option Judgment and Elimination

PROBLEM

=====

The expression $2 \cdot 3 \cdot 4 \cdot 5 + 1$ is equal to 121, since multiplication is carried out before addition. However, we can obtain values other than 121 for this expression if we are allowed to change it by inserting parentheses. For example, we can obtain 144 by writing $((2 \cdot 3 \cdot 4)) \cdot (5 + 1) = 144$.

\\]In total, how many values can be obtained from the expression $2 \cdot 3 \cdot 4 \cdot 5 + 1$ by inserting parentheses? (Note that rearranging terms is not allowed, o...

GROUND TRUTH ANSWER

=====

4

MODEL RESPONSE

=====

The expression $((2 \cdot 3 \cdot 4 \cdot 5 + 1))$ can yield distinct values only when parentheses are inserted such that the addition operation is performed before some of the multiplications. The distinct values arise from how many numbers are included in the product that is added to 1.

The possible values are:

- $((2 \cdot 3 \cdot 4 \cdot 5) + 1 = 120 + 1 = 121)$ (addition last)
- $((2 \cdot 3 \cdot 4 \cdot 5 + 1) = 2 \cdot (60 + 1) = 2 \cdot 61 = 122)$
- $((2 \cdot 3 \cdot 4 \cdot 5 + 1) = 2 \cdot 3 \cdot (20 + 1) = 2 \cdot 3 \cdot 21 = 126)$
- $((2 \cdot 3 \cdot 4 \cdot 5 + 1) = 24 \cdot 6 = 144)$

Any other parenthesization, such as grouping only the multiplications, results in the same value as one of the above due to the associativity of multi...

CHAIN OF THOUGHT (First 1500 chars)

=====

First, the expression is $((2 \cdot 3 \cdot 4 \cdot 5 + 1))$. Without parentheses, it's evaluated as $((2 \cdot 3 \cdot 4 \cdot 5) + 1 = 120 + 1 = 121)$. But we can insert parentheses to change the order of operations.

The goal is to find how many distinct values we can get by inserting parentheses in different ways. We're not allowed to rearrange the terms, only add parentheses.

The expression has multiplication and addition, and we need to consider the precedence. Multiplication has higher precedence than addition, but parentheses can override that.

The expression has four numbers multiplied together and then added to 1. But with parentheses, we can group the operations differently.

Let me denote the expression as: $A \cdot B \cdot C \cdot D + E$, where $A=2$, $B=3$, $C=4$, $D=5$, $E=1$. But E is added at the end, so it's like $((A \cdot B \cdot C \cdot D + E))$.

Since we can insert parentheses, we need to consider all possible ways to parenthesize the multiplication and addition.

The expression involves both multiplication and addition, so we need to consider the structure where the addition can be done at different times.

Think of the expression as a sequence of operations: we have four multiplications and one addition. But since multiplication is associative, the way we group multiplications doesn't change the product, but when we add the 1, it might change things if we add it before some multiplications.

Ah, that's key. The addition of 1 can be incorporated at different poin...

Example for Strategy F: Reverse Thinking

PROBLEM

=====

A worker receives an annual wage of $\$20,000$, which he always deposits into a savings account at the end of the year. By the end of the third year (when he makes the third deposit), he wants to have at least $\$66,200$ in the account to finance the purchase of a house. What is the minimal compound interest rate that the savings account must provide? Express your answer as a percentage, but do not include the percent sign.

GROUND TRUTH ANSWER

=====

10

MODEL RESPONSE

=====

10

CHAIN OF THOUGHT (First 1500 chars)

=====

First, the worker deposits $\$20,000$ at the end of each year. He makes three deposits: one at the end of year 1, one at end of year 2, and one at end of year 3. By the end of year 3, after the third deposit, he wants at least $\$66,200$.

I need to find the minimal compound interest rate that will make this happen. Since it's compound interest, and deposits are made annually, I should consider the future value of each deposit.

Let the annual interest rate be r (as a decimal). So, for example, if r is 5%, then $r = 0.05$.

Now, let's think about the future value of each deposit at the end of year 3.

- The first deposit is made at the end of year 1. It will earn interest for 2 years (from end of year 1 to end of year 3). So, its future value is $20,000 * (1 + r)^2$.

- The second deposit is made at the end of year 2. It will earn interest for 1 year (from end of year 2 to end of year 3). So, its future value is $20,000 * (1 + r)^1$.

- The third deposit is made at the end of year 3. It doesn't earn any interest yet because it's deposited at the end, so its value is just $20,000$.

Therefore, the total amount at the end of year 3 is the sum of these three future values:

$$A = 20,000 * (1 + r)^2 + 20,000 * (1 + r) + 20,000$$

We want this amount to be at least $66,200$. Since we need the minimal rate, we can set it equal to $66,200$ for the minimum.

So:

$$20,000 * (1 + r)^2 + 20,000 * (1 + r) + 20,000 = 66,200$$

I can factor out $20,000$:

$$20,000 * [(1 + r)^2 + (1 + r) + 1] = 66,200$$

Now, divide ...

Example for Strategy G: Divergent and Convergent Thinking

PROBLEM

Given that a, b, c, d, e are real numbers such that

$$\begin{aligned} a + b + c + d + e &= 8, \\ a^2 + b^2 + c^2 + d^2 + e^2 &= 16. \end{aligned}$$

Determine the maximum value of e .

GROUND TRUTH ANSWER

$\frac{16}{5}$

MODEL RESPONSE

The maximum value of e is achieved when $a = b = c = d = \frac{6}{5}$ and $e = \frac{16}{5}$, satisfying the given conditions:

$$\begin{aligned} a + b + c + d + e &= 4 \cdot \frac{6}{5} + \frac{16}{5} = \frac{24}{5} + \frac{16}{5} = \frac{40}{5} = 8 \\ \text{and} \\ a^2 + b^2 + c^2 + d^2 + e^2 &= 4 \cdot \left(\frac{6}{5}\right)^2 + \left(\frac{16}{5}\right)^2 = 4 \cdot \frac{36}{25} + \frac{256}{25} = \frac{144}{25} + \frac{256}{25} = \frac{400}{25} = 16. \end{aligned}$$

The inequality derived from the constraints shows that e must satisfy $0 \leq e \leq \frac{16}{5}$. For $e > \frac{16}{5}$, the sum of squares exceeds 16, which violates the condition. Thus, $\frac{16}{5}$ is the maximum value of e .

$$\boxed{\frac{16}{5}}$$

CHAIN OF THOUGHT (First 1500 chars)

First, I need to find the maximum value of e given that the sum of a, b, c, d, e is 8 and the sum of their squares is 16. Since a, b, c, d are real numbers, I can think about how to maximize e while satisfying these conditions.

I recall that for a fixed sum, the sum of squares is minimized when all values are equal, but here I want to maximize e , so probably I need to make a, b, c, d as small as possible in terms of their squares, but since the sum is fixed, I need to balance that.

Let me denote $S = a + b + c + d + e = 8$, and $Q = a^2 + b^2 + c^2 + d^2 + e^2 = 16$.

I want to maximize e . So, for fixed e , I need to minimize the sum of squares of a, b, c, d subject to their sum being $S - e = 8 - e$.

Let $T = a + b + c + d = 8 - e$.

And I need to minimize $a^2 + b^2 + c^2 + d^2$ given that their sum is T .

I know that for a fixed sum, the sum of squares is minimized when all values are equal, so min of $a^2 + b^2 + c^2 + d^2$ is when $a = b = c = d = T/4$, so min sum of squares is $4 \cdot (T/4)^2 = T^2 / 4$.

But in this case, I have the constraint that the total sum of squares is 16, so for given e , the sum of squares for a, b, c, d must be at least the minimum, but since I want to achieve $Q=16$, I need that the sum of squares for a, b, c, d is exactly $Q - e^2 = 16 - e^2$.

So, for it to be possible, I need that $16 - e^2 \geq \text{min sum of squares for } a, b, c, d$, which is $T^2 / 4$, where $T = 8 - e$.

$$\text{So, } 16 - e^2 \geq (8 - e)^2 / 4$$

That is the inequality that must hold.

And since I want to maximize e , I proba...

Example for Strategy H: Counterexample Testing

PROBLEM

What is the least positive integer multiple of 30 that can be written with only the digits 0 and 2?

GROUND TRUTH ANSWER

2220

MODEL RESPONSE

The least positive integer multiple of 30 that can be written using only the digits 0 and 2 is 2220.

- 2220 is a multiple of 30, as $30 \times 74 = 2220$.
- It is written with only the digits 2 and 0.
- It is the smallest such number, as no smaller number (e.g., 20, 200, 220, 2000, 2020, 2200) meets both criteria: ending with 0 (for divisibility by 10) and having a digit sum divisible by 3 (for divisibility by 3). The digit sum of 2220 is $2 + 2 + 2 + 0 = 6$, which is divisible by 3.

Thus, 2220 is the solution.

CHAIN OF THOUGHT (First 1500 chars)

First, the question is: What is the least positive integer multiple of 30 that can be written with only the digits 0 and 2? So, I need to find a number that is a multiple of 30, and when written in decimal, it only contains the digits 0 and 2.

Since it's a multiple of 30, it must be divisible by both 10 and 3. Because $30 = 10 \times 3$.

Divisible by 10 means that the number must end with a 0. So, the last digit must be 0.

Divisible by 3 means that the sum of its digits must be divisible by 3.

The number only has digits 0 and 2. So, let the number have k digits, with only 0s and 2s, and it ends with 0.

Let S be the sum of the digits. Since the digits are only 0 and 2, S must be a multiple of 3 because of the divisibility rule for 3.

S is the sum of some 2s and 0s, so $S = 2 \times n$, where n is the number of 2s in the number.

So, $S = 2n$ must be divisible by 3. Therefore, $2n \equiv 0 \pmod{3}$, which means n must be divisible by 3 because 2 and 3 are coprime. So n is a multiple of 3.

So, the number must have a multiple of 3 number of 2s, and it must end with a 0.

Now, I need the least such multiple, so I should find the smallest number that meets these criteria.

Since it has to be a multiple of 30, and it ends with 0, I need to consider numbers that are multiples of 30 with only digits 0 and 2.

I should start with small numbers and work my way up.

The number must end with 0, so let's think about the possible numbers.

The smallest number with only 0 and 2 that ends with 0 is 20, but 20 ...

Example for Strategy I: Association Method

PROBLEM

Compute

$$\frac{1}{\cos^2 10^\circ} + \frac{1}{\sin^2 20^\circ} + \frac{1}{\sin^2 40^\circ}.$$

GROUND TRUTH ANSWER

12

MODEL RESPONSE

The expression to compute is $\frac{1}{\cos^2 10^\circ} + \frac{1}{\sin^2 20^\circ} + \frac{1}{\sin^2 40^\circ}$.

Note that $\cos 10^\circ = \sin 80^\circ$, so $\frac{1}{\cos^2 10^\circ} = \frac{1}{\sin^2 80^\circ}$. Thus, the expression becomes:

$$\frac{1}{\sin^2 80^\circ} + \frac{1}{\sin^2 20^\circ} + \frac{1}{\sin^2 40^\circ}.$$

This matches the form $\frac{1}{\sin^2 \theta} + \frac{1}{\sin^2 (60^\circ - \theta)} + \frac{1}{\sin^2 (60^\circ + \theta)}$ for $\theta = 20^\circ$, since:

- $60^\circ - 20^\circ = 40^\circ$,
- $60^\circ + 20^\circ = 80^\circ$.

The identity states that:

$$\frac{1}{\sin^2 \theta} + \frac{1}{\sin^2 (60^\circ - \theta)} + \frac{1}{\sin^2 (60^\circ + \theta)} = 9 \csc^2 \theta.$$

CHAIN OF THOUGHT (First 1500 chars)

First, I need to compute this expression: $\frac{1}{\cos^2 10^\circ} + \frac{1}{\sin^2 20^\circ} + \frac{1}{\sin^2 40^\circ}$. Since it involves trigonometric functions, I should look for identities or properties that might simplify this.

I recall that $\frac{1}{\cos^2 \theta} = \sec^2 \theta$ and $\frac{1}{\sin^2 \theta} = \csc^2 \theta$, but that might not help directly. Also, I know that $\sec^2 \theta = 1 + \tan^2 \theta$ and $\csc^2 \theta = 1 + \cot^2 \theta$, but again, that might complicate things.

Perhaps I can use the fact that these angles are related: 10° , 20° , 40° . Notice that 20° is double 10° , and 40° is double 20° . So, maybe I can use double-angle formulas.

Let me denote the expression as $E = \frac{1}{\cos^2 10^\circ} + \frac{1}{\sin^2 20^\circ} + \frac{1}{\sin^2 40^\circ}$.

I remember that in trigonometry, there are identities for sums like this. Maybe I can find a common denominator or something.

Another thought: perhaps I can use the identity for $\frac{1}{\sin^2 \theta}$ in terms of cosecant, but let's think differently.

I know that $\sin 2\theta = 2 \sin \theta \cos \theta$, so for example, $\sin 20^\circ = 2 \sin 10^\circ \cos 10^\circ$.

Similarly, $\sin 40^\circ = 2 \sin 20^\circ \cos 20^\circ$.

But I have $\frac{1}{\sin^2 20^\circ}$ and $\frac{1}{\sin^2 40^\circ}$, so let's see if I can express them in terms of smaller angles.

Consid...